SIMATS ENGINEERING
ASSESSMENT TOOL 2

Course Name: MLA03

Course Code: Reinforcement Learning for Environment and Agent

Course Outcome: CO1 — Analyze reinforcement learning frameworks and Markov Decision Process concepts to model decision-making problems (BL2, BL3)

Faculty: Dr Dumpala Prasanth

Questions: 10 Total Marks: 50 (5 marks each)

Question 1: RL-Based Warehouse Robot

A logistics company is developing an RL-based warehouse robot that learns to pick, pack, and transport items efficiently while avoiding obstacles and minimizing delays. Analyze how the components of a Markov Decision Process (states, actions, rewards, policy, and environment) can be defined for this system. Design a reward function that balances speed, accuracy, and safety, and evaluate how reward design influences the robot's learning and behavior. (5 Marks)

Answer:

The warehouse robot task is modeled as a Markov Decision Process (MDP) in which the robot (agent) interacts with the warehouse (environment) to learn efficient picking, packing, and transport behavior.

MDP Formulation

- **States (S):** Robot's current location (grid coordinates), orientation, battery level, item(s) currently held, positions of obstacles/other robots, and status of the target shelf/packing station.
- **Actions (A):** Move forward/backward/left/right, rotate, pick item, place item, pack item, charge battery, and wait/stop.
- **Rewards (R):** Numerical feedback returned after each action, shaped to reflect task success, speed, and safety.
- **Policy (π):** A mapping from states to actions (e.g., a learned neural-network policy) that decides the robot's next move to maximize expected cumulative reward.
- **Environment:** The warehouse floor plan, shelving layout, moving obstacles (other robots/humans), and the order-management system that issues pick-pack tasks.

Reward Function Design

- Positive reward (+10) for successfully picking the correct item.
- Positive reward (+15) for correctly packing/delivering an item to the right location.
- Small negative reward (-0.1) per time step to encourage speed and discourage idling.
- Large negative reward (-20) for collisions with obstacles, shelves, or other robots (safety penalty).
- Negative reward (-5) for picking the wrong item (accuracy penalty).
- Bonus reward for completing an order within a target time window.

Analysis & Evaluation

- A reward function that overweights speed (large time penalty) can cause the robot to rush and increase collisions, harming safety.
- Overweighting safety (very large collision penalty) may make the robot overly cautious, reducing throughput.

- Balanced, multi-objective shaping — combining task-completion rewards, small time penalties, and strong but not overwhelming safety penalties — produces an agent that is efficient yet careful.
- Sparse rewards (only at task completion) slow learning, so intermediate shaping rewards (for correct picks) accelerate convergence.
- Reward design directly shapes emergent behavior: the agent optimizes exactly what is rewarded, so poorly designed rewards can lead to unintended shortcuts (e.g., dropping items to save time).

Question 2: RL-Based Movie Recommendation System

A streaming platform wants to use Reinforcement Learning to recommend movies dynamically based on user preferences and viewing history. Apply RL concepts to construct a suitable state representation, action space, and reward function. Analyze how different reward signals affect user engagement and evaluate the trade-off between exploration and exploitation. (5 Marks)

Answer:

The recommendation problem is modeled as a sequential decision process where the streaming platform (agent) selects content to show a user (environment) to maximize long-term engagement.

MDP Formulation

- **State (S):** User profile (demographics, preferences), recent viewing history, watch duration/completion rate, genre affinities, time of day, and device type — typically encoded as a feature vector or embedding.
- **Action (A):** Recommending a specific movie/show, or a ranked list/slate of titles, from the available catalog.
- **Reward (R):** Signal derived from user response to the recommendation (click, watch time, completion, rating, or subscription retention).
- **Policy (π):** A model (e.g., contextual bandit or deep RL policy) mapping user state to a recommendation or ranked slate.

Reward Function Design

- +1 for a click on the recommended title.
- Proportional reward based on percentage of the video watched (e.g., completion rate).
- Bonus reward for explicit positive feedback (thumbs-up/high rating).
- Negative reward for a quick abandonment (watching <10% before stopping).
- Delayed reward tied to session length or next-session return, capturing long-term engagement rather than a single click.

Analysis & Evaluation

- Reward signals based only on clicks can encourage clickbait-style recommendations (click-through without genuine satisfaction).
- Watch-time or completion-based rewards better reflect true engagement but are noisier and delayed, complicating credit assignment.
- Exploitation (recommending known favorites) maximizes short-term reward but risks filter bubbles and stale recommendations.
- Exploration (recommending novel/diverse content) risks short-term dissatisfaction but improves long-term catalog coverage and discovers new preferences.
- A common strategy is epsilon-greedy or upper-confidence-bound exploration layered on top of a primarily exploitative policy, gradually reducing exploration as user preferences become well estimated.

Question 3: RL for Agricultural Irrigation and Fertilizer Optimization

An autonomous agricultural system uses RL to optimize irrigation and fertilizer usage based on soil conditions and weather forecasts. Design an RL framework by defining states, actions, and rewards. Analyze the effect of delayed rewards on crop yield optimization and justify the choice of a suitable learning algorithm. (5 Marks)

Answer:

The autonomous agricultural system is framed as an MDP where the agent controls irrigation and fertilizer application to optimize crop yield while conserving resources.

MDP Formulation

- **States (S):** Soil moisture, soil nutrient levels (N-P-K), crop growth stage, weather forecast (rainfall, temperature, humidity), and historical yield data.
- **Actions (A):** Amount of water to irrigate, amount/type of fertilizer to apply, or no action (wait), typically discretized into levels (none/low/medium/high).
- **Rewards (R):** Feedback tied to crop health indicators, resource efficiency, and — ultimately — yield at harvest.

Reward Function Design

- Positive reward proportional to improvement in crop health indicators (e.g., NDVI/vegetation index) after each action.
- Negative reward for water/fertilizer overuse (resource cost penalty) to encourage sustainability.
- Large positive reward at harvest time proportional to final crop yield.
- Penalty for under-watering leading to plant stress or nutrient deficiency symptoms.

Analysis & Evaluation

- Crop yield is only observable at harvest, so the true reward is heavily delayed relative to daily irrigation/fertilizer decisions, creating a difficult credit-assignment problem.
- Delayed rewards require algorithms that reason over long horizons and can propagate final yield outcomes back to earlier state-action pairs (e.g., discounted returns with an appropriately high discount factor).
- Model-based RL or algorithms like Proximal Policy Optimization (PPO) with intermediate shaping rewards are well suited here, since they can incorporate short-term health signals to reduce the effective delay while still optimizing for the long-term yield objective.
- Off-policy methods (e.g., DDPG for continuous water/fertilizer amounts) are useful given the continuous action space of irrigation volume.

Question 4: RL for Network Intrusion Detection and Response

A cybersecurity system uses RL to detect and respond to network attacks in real time. Analyze how states and environment can be modeled for intrusion detection. Design appropriate actions and reward functions, and evaluate the challenges of sparse rewards and real-time decision-making. (5 Marks)

Answer:

The cybersecurity system is modeled as an MDP in which the agent monitors live network traffic and takes defensive actions to detect and mitigate attacks.

MDP Formulation

- **States (S):** Network traffic features (packet rate, protocol type, source/destination IP patterns, connection duration), system logs, and current threat-alert level.
- **Actions (A):** Flag traffic as suspicious, block an IP/connection, throttle bandwidth, isolate a host, escalate to a human analyst, or take no action.

- **Environment:** The live network infrastructure, generating a continuous, high-volume, non-stationary stream of traffic that includes both benign and malicious patterns.

Reward Function Design

- Large positive reward for correctly identifying and blocking a genuine attack (true positive).
- Large negative reward for missing an actual attack (false negative), reflecting its high cost.
- Moderate negative reward for false positives (blocking legitimate traffic), since this disrupts normal operations.
- Small negative reward per time step during unresolved threats to encourage fast response.

Analysis & Evaluation

- Malicious events are rare relative to normal traffic, producing sparse and imbalanced rewards, which makes it hard for the agent to gather enough positive learning signal.
- Real-time decision-making requires the agent to act within tight latency constraints, limiting the use of computationally expensive planning methods.
- Sparse rewards can be mitigated using reward shaping (e.g., partial credit for correctly flagging suspicious-but-unconfirmed behavior) or curriculum learning with simulated attack scenarios.
- The evolving nature of attacks (concept drift) means the environment is non-stationary, requiring continual learning or periodic retraining to remain effective.

Question 5: RL for Ride-Sharing Driver Allocation and Dynamic Pricing

A ride-sharing company plans to use RL to optimize driver allocation and dynamic pricing strategies. Apply RL concepts to define states, actions, and rewards. Analyze how environmental uncertainty affects learning and evaluate ethical concerns related to pricing strategies. (5 Marks)

Answer:

The ride-sharing problem is modeled as an MDP in which the platform allocates drivers and sets prices in response to fluctuating demand and supply across a city.

MDP Formulation

- **States (S):** Current driver locations, rider demand density by zone, time of day, weather, traffic conditions, and historical demand patterns.
- **Actions (A):** Dispatch a specific driver to a rider, reposition idle drivers to high-demand zones, and set a price multiplier (surge pricing level) for a zone.
- **Rewards (R):** Feedback based on completed trips, revenue, wait times, and overall marketplace efficiency.

Reward Function Design

- Positive reward for successfully matching a rider with a driver quickly (short wait time).
- Revenue-based reward proportional to fare collected, encouraging efficient pricing.
- Negative reward for long rider wait times or unmatched ride requests (lost demand).
- Penalty for excessive surge pricing that leads to rider cancellations, balancing revenue against fairness.

Analysis & Evaluation

- Demand and traffic are highly uncertain and time-varying, so the agent must continuously adapt; this environmental uncertainty makes long-horizon planning difficult and favors online/adaptive learning methods.
- Excessive surge pricing during emergencies or for vulnerable riders raises fairness and ethical concerns, as pricing purely for profit can disproportionately burden certain users.

- Ethical deployment requires constraints or fairness-aware reward terms (e.g., capping surge multipliers, ensuring equitable service across neighborhoods) rather than optimizing revenue alone.
- Transparency in how prices are set is important to maintain rider trust and comply with regulatory expectations around dynamic pricing.

Question 6: RL for Smart Home Automation

A smart home automation system uses RL to control lighting, heating, and cooling based on occupancy and weather conditions. Design an RL framework including states, actions, and reward function. Analyze how conflicting objectives like comfort and energy saving can be handled and evaluate the impact of reward shaping. (5 Marks)

Answer:

The smart home system is modeled as an MDP where the agent controls lighting, heating, and cooling to balance occupant comfort with energy efficiency.

MDP Formulation

- **States (S):** Occupancy status, indoor temperature/humidity, outdoor weather, time of day, and current device settings (thermostat level, light brightness).
- **Actions (A):** Adjust thermostat setpoint, turn lights on/off or dim them, switch HVAC modes, or maintain current settings.
- **Rewards (R):** Feedback combining comfort satisfaction and energy savings.

Reward Function Design

- Positive reward for maintaining temperature/lighting within the occupant's comfort range.
- Positive reward proportional to energy saved compared to a baseline (e.g., always-on) policy.
- Negative reward for large or frequent deviations from comfort preferences (discomfort penalty).
- Negative reward for excessive energy consumption during unoccupied periods.

Analysis & Evaluation

- Comfort and energy savings are often conflicting objectives — maximum comfort (constant heating/cooling) tends to maximize energy use, while minimizing energy use tends to reduce comfort.
- This is typically handled with a weighted multi-objective reward (e.g., $R = w1 \cdot comfort - w2 \cdot energy\_cost$), where weights are tuned to reflect user priorities or adjusted dynamically based on occupancy.
- Reward shaping (adding intermediate signals like rate-of-change penalties for temperature swings) improves learning stability and produces smoother, more predictable control.
- Poorly tuned shaping can cause the agent to exploit unintended shortcuts (e.g., ignoring comfort during 'unoccupied' misclassification), so careful validation against real occupant feedback is needed.

Question 7: RL Agent for a Complex Strategy Game

A gaming company is developing an RL-based agent to play a complex strategy game against human players. Analyze how the problem can be modeled as an MDP. Design a reward structure that encourages long-term strategy and evaluate how exploration strategies affect performance. (5 Marks)

Answer:

The strategy game is modeled as an MDP (or a Markov Game in the multi-agent/adversarial setting) in which the agent selects moves to defeat a human opponent over an extended sequence of turns.

MDP Formulation

- **States (S):** Current game board/map configuration, resources, unit positions, opponent's visible actions, and turn number.

- **Actions (A):** The set of legal moves available at a given state (unit movement, resource allocation, attack, build, or defend decisions).
- **Rewards (R):** Primarily sparse — win/loss/draw at the end of the game — supplemented with intermediate shaping signals.

Reward Function Design

- Large positive reward (+1) for winning the game; large negative reward (-1) for losing; small/neutral reward for a draw.
- Small intermediate rewards for strategic sub-goals (capturing territory, resource accumulation, eliminating enemy units).
- Small negative reward for wasteful or clearly suboptimal actions to discourage inefficient play.

Analysis & Evaluation

- Rewarding only the final win/loss encourages genuine long-term strategy but suffers from a very sparse and delayed signal, making learning slow, especially in long games.
- Adding intermediate shaping rewards speeds up learning but risks the agent optimizing short-term sub-goals (e.g., hoarding resources) at the expense of the actual win condition — a form of reward hacking.
- Exploration strategies (e.g., self-play with an evolving opponent, entropy-regularized policies, or Monte Carlo Tree Search-guided exploration) are critical: too little exploration causes the agent to get stuck in a narrow strategy that a skilled human can counter; too much exploration slows convergence to strong play.
- Techniques like curriculum learning (increasing opponent difficulty over time) help balance exploration and exploitation as the agent's skill improves.

Question 8: RL for Real-Time Fraudulent Transaction Detection

A financial institution uses RL to detect fraudulent transactions in real time. Apply RL principles to define the state space, action space, and reward function. Analyze the challenges of imbalanced data and delayed feedback, and evaluate the risks associated with incorrect decisions. (5 Marks)

Answer:

The fraud-detection problem is modeled as an MDP in which the agent evaluates each incoming transaction and decides how to act to minimize financial loss while maintaining customer experience.

MDP Formulation

- **States (S):** Transaction features (amount, location, merchant category, time), account history, recent transaction patterns, and device/behavioral fingerprint.
- **Actions (A):** Approve the transaction, decline it, flag for manual review, or request additional authentication (e.g., OTP verification).
- **Rewards (R):** Feedback reflecting the correctness of the decision, weighted by the financial and reputational cost of errors.

Reward Function Design

- Large positive reward for correctly blocking a fraudulent transaction (true positive).
- Large negative reward for approving a fraudulent transaction (false negative) — reflecting direct financial loss.
- Moderate negative reward for declining a legitimate transaction (false positive) — reflecting customer dissatisfaction and lost business.
- Small positive reward for correctly approving legitimate transactions with minimal friction.

Analysis & Evaluation

- Fraudulent transactions are rare compared to legitimate ones, creating a highly imbalanced dataset that biases naive learning toward always predicting 'legitimate.'
- Confirmation of fraud is often delayed (customer disputes may surface days or weeks later), so the true reward signal for a decision is not immediately available — a delayed feedback problem similar to Question 3.
- Techniques such as reward weighting for the minority (fraud) class, synthetic oversampling, or anomaly-detection pretraining can help address class imbalance.
- Incorrect decisions carry asymmetric risk: false negatives cause direct monetary loss, while false positives damage customer trust — so the reward function and decision threshold must be carefully calibrated rather than optimizing raw accuracy.

Question 9: RL for Personalized Rehabilitation Exercise Recommendation

A healthcare system uses RL to recommend personalized rehabilitation exercises for patients based on their progress. Design an RL framework with suitable states, actions, and rewards. Analyze the impact of delayed rewards and patient variability, and evaluate ethical concerns and safety considerations. (5 Marks)

Answer:

The healthcare rehabilitation system is modeled as an MDP in which the agent selects exercises tailored to a patient's evolving recovery status.

MDP Formulation

- **States (S):** Patient's current mobility/strength metrics, pain levels, exercise completion history, adherence rate, and medical constraints.
- **Actions (A):** Recommend a specific exercise, adjust exercise intensity/duration, or suggest rest, from a clinically approved set of options.
- **Rewards (R):** Feedback tied to measurable recovery progress and patient safety/comfort.

Reward Function Design

- Positive reward for measurable improvement in mobility/strength after an exercise session.
- Positive reward for high patient adherence and completion of recommended exercises.
- Large negative reward for exercises that cause pain spikes or risk of injury.
- Negative reward for recommendations that lead to reduced adherence (e.g., overly difficult routines causing dropout).

Analysis & Evaluation

- Recovery outcomes unfold over weeks, so rewards are delayed relative to individual exercise sessions, complicating attribution of which specific recommendations drove improvement.
- Patient variability (differing injury types, ages, pain tolerance, comorbidities) means a single policy may not generalize well; personalization or patient-clustering approaches are typically needed.
- Safety is paramount: because incorrect recommendations can cause physical harm, the system must incorporate conservative action constraints, clinician oversight, and possibly offline/behavior-constrained RL trained on historical clinical data rather than pure online exploration.
- Ethical considerations include informed consent, avoiding over-reliance on automated recommendations for high-risk patients, and ensuring transparency so clinicians can override the agent's suggestions.

Question 10: RL for Smart Grid Electricity Distribution

A smart grid system uses RL to manage electricity distribution and reduce peak load demand. Analyze how the components of an MDP can be defined in this context. Design a reward function to balance efficiency and reliability, and evaluate the challenges of scalability and real-time learning. (5 Marks)

Answer:

The smart grid system is modeled as an MDP in which the agent manages electricity distribution and demand response to balance supply, cost, and reliability.

MDP Formulation

- **States (S):** Current electricity demand across regions, generation capacity/mix (renewable and conventional), grid load levels, weather (affecting renewable output), and storage/battery levels.
- **Actions (A):** Adjust power distribution/routing between regions, activate or deactivate generation sources, charge/discharge storage, or trigger demand-response signals (e.g., dynamic pricing to shift consumption).
- **Rewards (R):** Feedback balancing cost efficiency, grid stability, and reliability.

Reward Function Design

- Positive reward for successfully meeting demand without shortages or blackouts.
- Positive reward proportional to reduction in peak load and efficient use of renewable energy sources.
- Large negative reward for blackouts, voltage instability, or equipment overload events.
- Negative reward for excessive reliance on costly backup generation during peak periods.

Analysis & Evaluation

- Balancing efficiency and reliability requires the reward to penalize instability far more heavily than modest inefficiency, since blackouts carry severe real-world consequences.
- Scalability is a major challenge: a real grid involves many interconnected regions and devices, leading to a very large state-action space that single-agent RL may not handle well — multi-agent or hierarchical RL approaches are often needed.
- Real-time learning is constrained by the grid's need for near-instant, highly reliable decisions, limiting the use of slow-converging or unstable learning algorithms; techniques like safe RL (with hard operational constraints) are commonly applied.
- Non-stationarity from weather-dependent renewable generation and changing consumption patterns requires the agent to continually adapt, favoring online or continually-retrained policies over static ones.

Code of Conduct

I _____S. Daksha Bala Vardhan.(192325126).________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____S. Daksha Bala Vardhan ________________________

Rubrics for Case Study

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Begin
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understand identified correc
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply re
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no anal reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no s
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; dif understand